Introduction to Databases

A database is an organized collection of structured data that is stored and managed electronically. Databases are essential tools for managing, storing, and retrieving data efficiently, and they play a vital role in modern applications.

Database Management Systems (DBMS)

A Database Management System (DBMS) is software that interacts with users, applications, and the database itself to capture, store, and manage data. It provides an interface for performing various operations on the data, such as inserting, updating, deleting, and retrieving data. The primary goal of a DBMS is to ensure that data remains consistent, secure, and easily accessible.

There are two main types of DBMSs:

- **Relational Database Management Systems (RDBMS)**: These systems store data in tables with predefined relationships between them. The most common query language for RDBMSs is SQL (Structured Query Language).
- **Non-Relational Database Management Systems (NoSQL)**: These systems store data in various formats, such as key-value, document, column-family, or graph. NoSQL databases are known for their ability to scale horizontally and handle unstructured or semi-structured data.

Overview of SQL and NoSQL databases

Databases can be broadly classified into two categories: SQL (Structured Query Language) and NoSQL (Not only SQL) databases. SQL databases, also known as relational databases, are based on the relational model, where data is stored in tables with predefined schema and relationships between them. Some popular SQL databases include MySQL, PostgreSQL, Microsoft SQL Server, and Oracle. SQL databases are known for their consistency, reliability, and powerful query capabilities.

On the other hand, NoSQL databases are a diverse group of non-relational databases that prioritize flexibility, scalability, and performance under specific workloads. NoSQL databases can be further categorized into document databases, key-value stores, column-family stores, and graph databases, each with their unique data models and use cases. Some widely-used NoSQL databases are MongoDB, Redis, Apache Cassandra, and Neo4j.

High level differences between SQL and NoSQL

1. Storage: SQL stores data in tables where each row represents an entity and each column represents a data point about that entity; for example, if we are storing a car entity in a table, different columns could be ‘Color’, ‘Make’, ‘Model’, and so on.

NoSQL databases have different data storage models. The main ones are key-value, document, graph, and columnar. We will discuss differences between these databases below.

2. Schema: In SQL, each record conforms to a fixed schema, meaning the columns must be decided and chosen before data entry and each row must have data for each column. The schema can be altered later, but it involves modifying the whole database and going offline.

In NoSQL, schemas are dynamic. Columns can be added on the fly and each ‘row’ (or equivalent) doesn’t have to contain data for each ‘column.'

3. Querying: SQL databases use SQL (structured query language) for defining and manipulating the data, which is very powerful. In a NoSQL database, queries are focused on a collection of documents. Sometimes it is also called UnQL (Unstructured Query Language). Different databases have different syntax for using UnQL.

4. Scalability: In most common situations, SQL databases are vertically scalable, i.e., by increasing the horsepower (higher Memory, CPU, etc.) of the hardware, which can get very expensive. It is possible to scale a relational database across multiple servers, but this is a challenging and time-consuming process.

On the other hand, NoSQL databases are horizontally scalable, meaning we can add more servers easily in our NoSQL database infrastructure to handle a lot of traffic. Any cheap commodity hardware or cloud instances can host NoSQL databases, thus making it a lot more cost-effective than vertical scaling. A lot of NoSQL technologies also distribute data across servers automatically.

5. Reliability or ACID Compliancy (Atomicity, Consistency, Isolation, Durability): The vast majority of relational databases are ACID compliant. So, when it comes to data reliability and safe guarantee of performing transactions, SQL databases are still the better bet.

Most of the NoSQL solutions sacrifice ACID compliance for availability, performance, and scalability.